Himalayan Expeditions: Summit Success

Data report for the SML project: loading the data, every variable described and assessed, problems and limitations, project ideas, and three course models with first results.

Source	The Himalayan Database (Elizabeth Hawley's records of all expeditions in the Nepal Himalaya), published as TidyTuesday 2020-09-22
Tables	members (76,519 x 21), expeditions (10,364 x 16), peaks (468 x 8)
Period	1905 to spring 2019; 90% of the rows are from 1980 or later
Target	success: did the climber reach the summit? 38.2% yes overall, 40.7% in the modelling population
Verdict	**Good to use.** Easy to load, clean, interpretable and with a clear story. It needs care with leakage, clustered data (members of one expedition) and confounding.

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1. Loading the data

The data is three CSV files on GitHub. No login or Kaggle account is needed and the columns arrive with the right types (numbers, TRUE/FALSE, dates). Downloading takes a few seconds.

```
library(readr); library(dplyr)
tt <- "https://raw.githubusercontent.com/rfordatascience/tidytuesday/main/data/2020/2020-09-22/"
members     <- read_csv(paste0(tt, "members.csv"))      # one row per person per expedition
expeditions <- read_csv(paste0(tt, "expeditions.csv"))  # one row per expedition
peaks       <- read_csv(paste0(tt, "peaks.csv"))        # one row per mountain
```

The tables link through `expedition_id` and `peak_id`. One trap: the id `KANG10101` is used for two different expeditions (1910 and 2010), so join the expeditions on id **and** year:

```
d <- members |>
  left_join(select(expeditions, expedition_id, year, team_size = members, hired_staff,
                   exp_oxygen = oxygen_used, trekking_agency, termination_reason),
            by = c("expedition_id", "year")) |>
  left_join(select(peaks, peak_id, height_metres, first_ascent_year), by = "peak_id")
```

How hard was it? Loading took five lines. The real work came after: deciding which variables are allowed (leakage), grouping messy categories (525 role names, 213 countries) and choosing the population. That work is the content of the pre-processing slide.

2. The three tables and every variable

Verdicts: **Target** what we predict, **Use** a predictor known before the summit push, **Leakage** only known after or during the climb, must not be a predictor, **Filter** used to choose rows, not as a predictor, **Key** id for joins and for splitting, **Drop** not useful.

2.1 members: 76,519 rows x 21 columns (one row per person per expedition)

Variable	Type	Missing	What it looks like	Verdict	What I think
`success`	logical	0%	38.2% TRUE	Target	Summit of the main peak or, if that was the goal, a sub-peak. Clean, no missing values.
`expedition_id`	text	0%	10,350 expeditions; up to 99 members each	Key	Needed for joins and to split train/test by expedition.
`member_id`	text	0%	unique per expedition	Drop	Not consistent across expeditions, so the same person cannot be followed over time.
`peak_id / peak_name`	text	0%	391 peaks; Everest 21,813 rows, Cho Oyu 8,890, Ama Dablam 8,406	Use	Group into the top 10 peaks + Other. Success ranges from 19% (Annapurna I) to 53% (Ama Dablam).
`year`	number	0%	1905-2019, median 2004	Use	Strong trend: about 20% success in the 1980s, 49% in the 2010s. Dips in 2014 (Everest avalanche) and 2015 (earthquake).
`season`	text	0%	Spring 49%, Autumn 47%, Winter 3%, Summer 1%	Use	Spring 41%, Autumn 36%, Summer 27%, Winter 24% success. One 'Unknown' row (drop).
`sex`	text	<0.1%	91% M, 9% F	Use	Men 39%, women 33%; probably confounded with role (almost all hired staff are men).
`age`	number	4.6%	7-85, median 36	Use	Clear decline: 46% success under 25, 24% over 60. Impute the median and add a 'missing' flag (missing ages have lower success).
`citizenship`	text	0%	213 values; Nepal 21%, USA 8%, Japan 8%, UK 7%	Use	Group into the top 15 + Other. Nepal is mostly hired Sherpas, so it overlaps with hired.
`expedition_role`	text	0%	525 values; Climber 58%, H-A Worker 19%, Leader 13%	Filter	Some roles never try to summit (BC Manager 0.3%, Journalist 0%, Exp Doctor 11%). Keep climbing roles only and group them: climber / leader / hired staff.
`hired`	logical	0%	20.6% TRUE	Use	Hired staff (mostly Sherpas) summit far more often: 64% vs 31%.
`highpoint_metres`	number	28.5%	3,800-8,850 m	Leakage	How high the person got. It is basically the outcome.
`solo`	logical	0%	121 people (0.2%)	Use	Too rare to matter, but harmless.
`oxygen_used`	logical	0%	23.8% TRUE; 78% vs 26% success	Leakage	Partly an outcome: oxygen is switched on high up, so climbers who turned back low almost never used it (Figure 3). Use the expedition-level variable instead.
`died`	logical	0%	1,106 deaths (1.4%)	Leakage	Happens during the climb. Could be the target of a separate project.
`death_cause`	text	98.6%	Avalanche 369, Fall 331, AMS 102, ...	Leakage	Only filled for people who died.
`death_height_metres`	number	98.6%	median 6,600 m	Leakage	Only filled for people who died.
`injured`	logical	0%	1,713 (2.2%)	Leakage	Happens during the climb.
`injury_type`	text	97.8%	Frostbite 599, AMS 415, Illness 257, ...	Leakage	Only for injured people.
`injury_height_metres`	number	98.7%	median 7,100 m	Leakage	Only for injured people.

2.2 expeditions: 10,364 rows x 16 columns (one row per expedition)

Variable	Type	Missing	What it looks like	Verdict	What I think
`expedition_id`	text	0%	10,363 unique (KANG10101 twice)	Key	Join on id and year because of the duplicate.
`peak_id / peak_name`	text	0%	same as in members	Drop	Already in members.

Variable	Type	Missing	What it looks like	Verdict	What I think
year / season	num / text	0%	same as in members	Key	Used as join key (year).
basecamp_date	date	10.6%	1921-2019	Drop	Could give 'start within the season', but season already covers most of it.
highpoint_date	date	6.3%	date of summit or highpoint	Leakage	Known only after the climb.
termination_date	date	23.0%	end of expedition	Leakage	Expedition length is an outcome.
termination_reason	text	0%	Success 54%, bad weather 13%, bad conditions 11%, illness 4%, ...	Filter	Leakage as a predictor, but use it to drop expeditions that never attempted the climb (did not attempt, did not reach base camp, rumoured, unknown).
highpoint_metres	number	4.0%	3,500-8,850 m	Leakage	Outcome of the whole expedition.
members	number	0%	0-99, median 5	Use	Team size (non-hired members). Skewed: use log. 48 expeditions have 0 (data error).
member_deaths	number	0%	mostly 0, max 10	Leakage	Happens during the climb.
hired_staff	number	0%	0-99, median 1	Use	Support above base camp. Staff per member is a good feature.
hired_staff_deaths	number	0%	mostly 0, max 11	Leakage	Happens during the climb.
oxygen_used	logical	0%	28.1% of expeditions	Use	The expedition's oxygen plan. The most important predictor, and much less leaky than the member version.
trekking_agency	text	15.2%	829 agencies; Asian Trekking 853	Use	Too many levels, so turned into 'commercial expedition yes/no'.

2.3 peaks: 468 rows x 8 columns (one row per mountain)

Variable	Type	Missing	What it looks like	Verdict	What I think
peak_id / peak_name	text	0%	468 peaks (391 appear in members)	Key	15 member rows have a peak missing from this table (dropped).
peak_alternative_name	text	47.6%	e.g. Sagarmatha for Everest	Drop	Not informative.
height_metres	number	0%	5,407-8,850 m, median 6,560	Use	Not monotone: 7,000-8,000 m peaks have the lowest success (16-28%), the 8,000 m peaks are higher (40%) because of commercial routes and oxygen.
climbing_status	text	0%	Climbed 341, Unclimbed 127	Drop	Status today, not at the time of the expedition.
first_ascent_year	number	28.2%	1930-2019 (one value '201' is a typo)	Use	Turned into 'peak already climbed before this expedition'.
first_ascent_country	text	28.2%	Japan, Nepal 61; Japan 43; ...	Drop	Not relevant for later climbers.
first_ascent_expedition_id	text	28.8%	link to expeditions	Drop	Not needed.

3. Descriptive figures

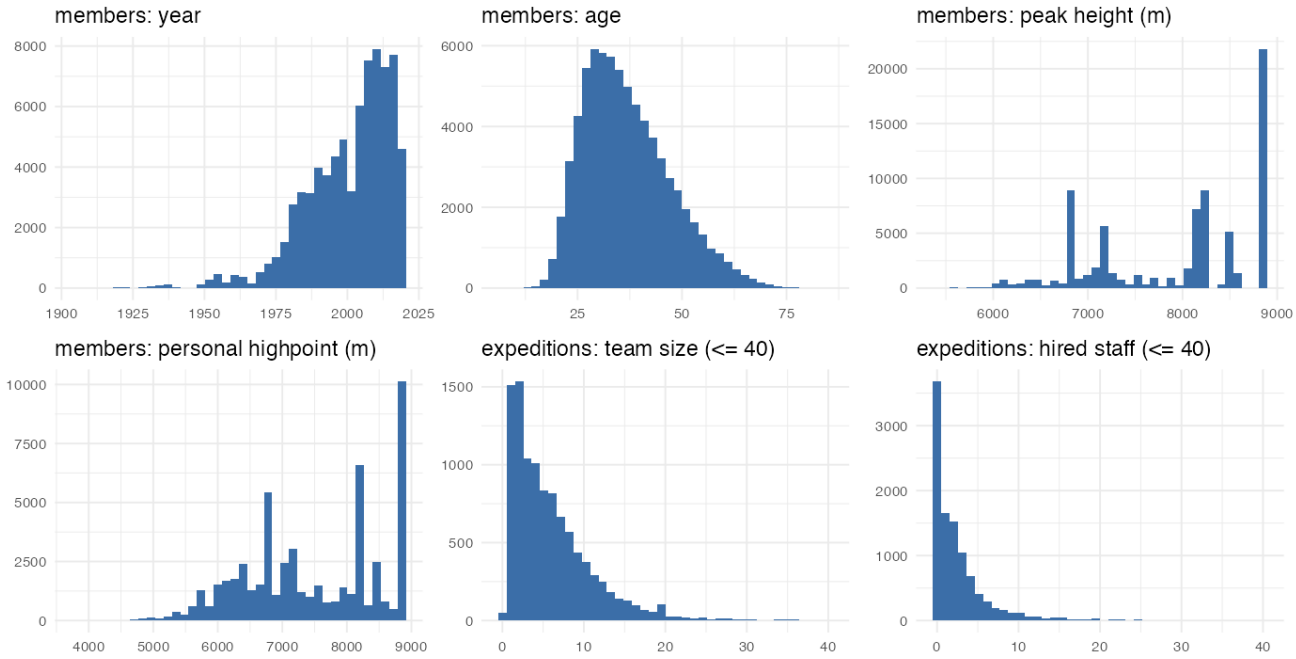


Figure 1. Distributions of the numeric variables. Most rows are recent; age is right-skewed; peak height has spikes at the popular peaks (Ama Dablam 6,812 m, Cho Oyu 8,188 m, Everest 8,850 m); team size and hired staff are very skewed.



Figure 2. Success rate by season, sex, hired and member-level oxygen (all members).

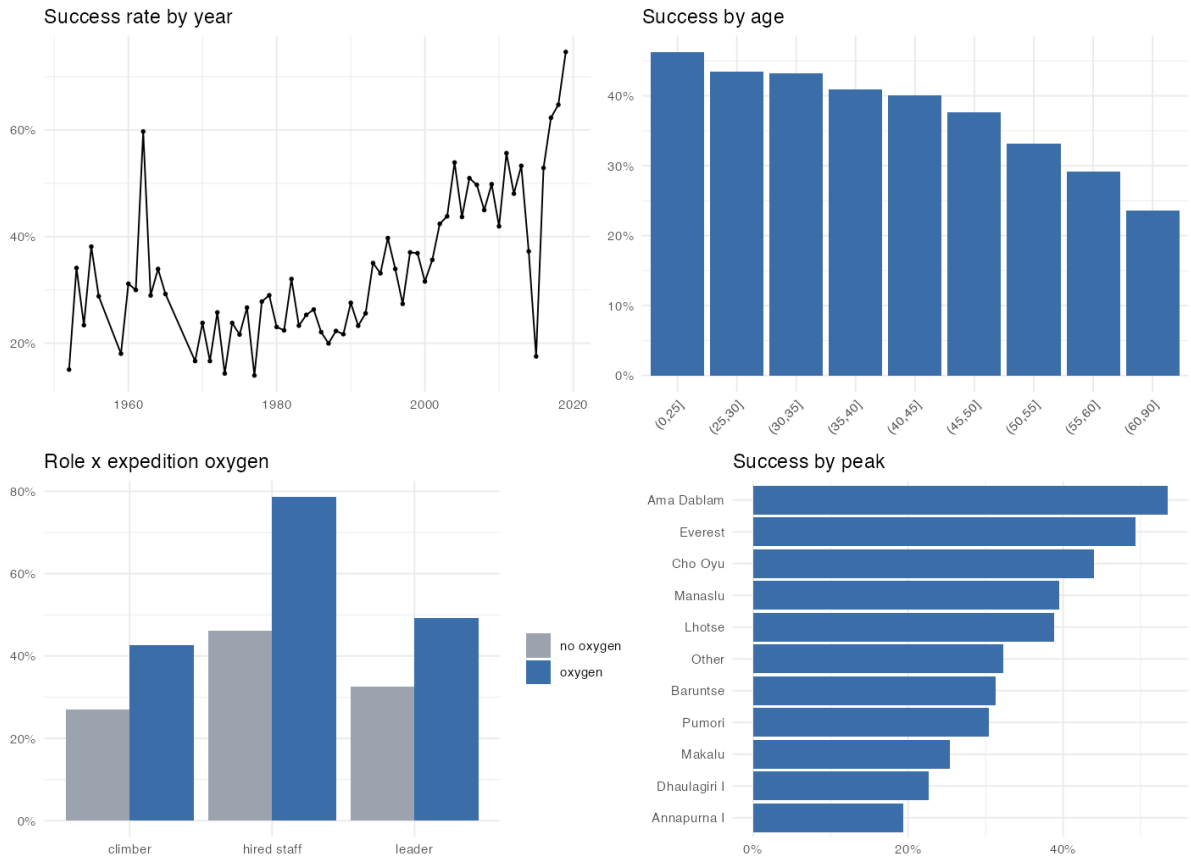


Figure 3a. Modelling population (climbing roles, 1950+): success by year, by age, by role x expedition oxygen and by peak.

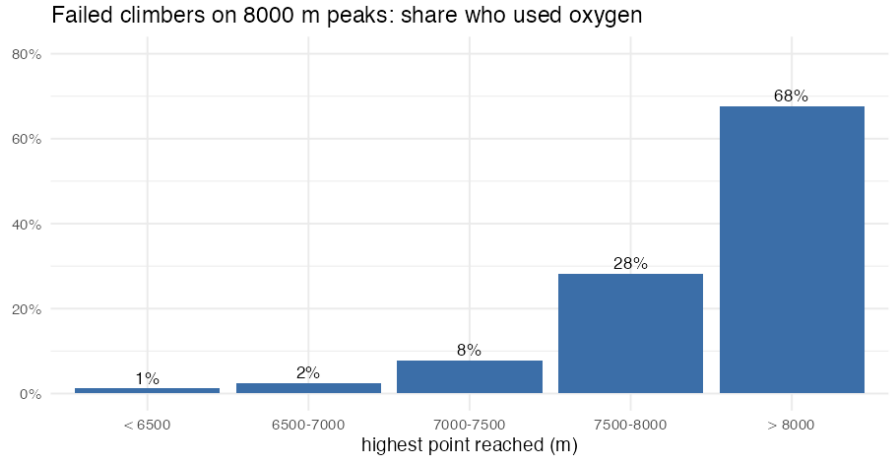


Figure 3b. Why member-level oxygen leaks: among climbers who failed on 8,000 m peaks, almost nobody who turned back below 7,000 m used oxygen, but 68% of those who failed above 8,000 m did.

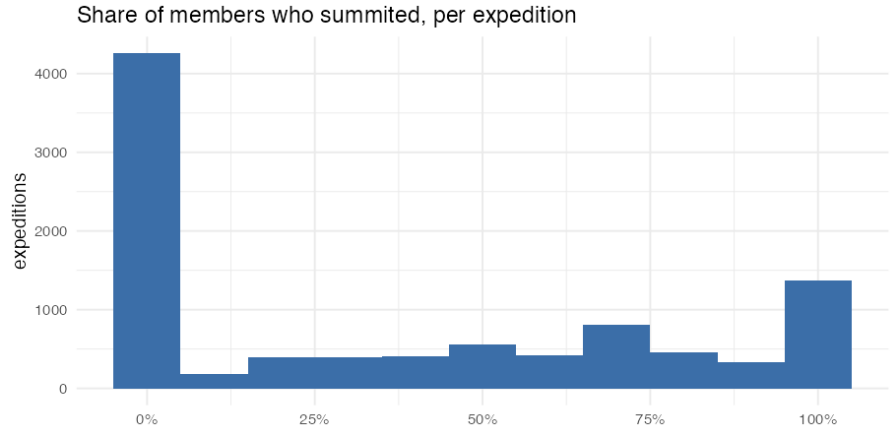


Figure 4. Within an expedition, members tend to share their fate: in 44% of expeditions everyone failed and in 14% everyone succeeded.

4. Problems and limitations

Problem	Explanation and how we handle it
Leakage	Many variables describe what happened on the mountain (highpoint, died, injured, dates, termination reason). The subtle one is member-level `oxygen_used`: with it, the random forest's AUC rises from 0.81 to 0.85, and that gain is fake. Use the expedition-level oxygen variable.
Weather is missing	Most failures are caused by weather or snow conditions (bad weather 13% and bad conditions 11% of expeditions), which are not in the data. This limits how good any model can be (AUC around 0.8) and should be said openly.
Clustered data	Members of one expedition share weather, route and team (Figure 4). A random split by row would put team-mates in both train and test and give too optimistic results, so the test split and the CV folds must be grouped by expedition.
Hidden leakage through season conditions	Even with a split by expedition, a test expedition on Everest in spring 2014 has training expeditions from the same avalanche season. A time-based test set (for example train up to 2016, test 2017-2019) is more honest and also answers a real question: can the past predict future seasons?
Confounding	Oxygen, Everest, commercial expeditions, hired Sherpas and recent years all go together. Partial dependence plots help to separate them, but the conclusions remain associations, not causal effects.
Who counts as a climber	Some members never tried to summit (base camp managers, doctors, journalists, film crews). They are easy to predict and would make the models look better than they are, so only climbing roles are kept.
Data errors	Duplicate expedition id KANG10101 (1910 and 2010), first_ascent_year = 201, 15 members with a peak missing from the peaks table, 48 expeditions with 0 members, one 7-year-old. All fixed or filtered.
Changing over time	Commercial climbing changed the Himalaya: success doubled since the 1990s. A model trained on old years may not fit recent ones; this is both a limitation and a possible story.
Coverage	Nepal Himalaya only (plus border peaks), up to spring 2019. People cannot be followed across expeditions, so climbing experience cannot be measured.

The modelling population after all filters: **70,353 climbers on 9,895 expeditions, 40.7% success** (climbing roles only, year 1950 or later, expeditions that actually attempted the climb, known peak, sex and season).

5. Project ideas

A plain 'predict success' project is a bit generic, and some findings are obvious (oxygen helps, older climbers do worse). A sharper headline question makes the presentation stronger.

A. Has the Himalaya become easier, or just more commercial? (recommended)

Success rose from about 20% to 49%. Is that explained by oxygen, Sherpa support and commercial expeditions, or is a time trend left after controlling for them? Fits partial dependence plots and the lasso coefficients well.

Strength	Risk
Clear story, uses the year trend, the plots carry the message.	Answers are associations, not causal.

B. Can the past predict future seasons?

Train on expeditions up to 2016 and test on 2017-2019. Fixes the hidden leakage and tests whether the models hold up over time.

Strength	Risk
Most honest evaluation; natural 'model assessment' discussion.	Recent years have higher success, so calibration may drift.

C. What is oxygen really worth?

Compare climbers with and without an oxygen plan at the same peak, age and year; show how the naive 78% vs 26% difference shrinks.

Strength	Risk
Shows confounding and leakage in one example.	Close to a causal question that the data cannot fully answer.

D. Who dies in the Himalaya?

Different target: died (1.4%). Very imbalanced, with real cost asymmetry (loss matrix, thresholds).

Strength	Risk
Strong motivation, imbalanced-class methods.	Only 1,106 deaths and very noisy; harder to get good results.

Recommendation: combine A and B. Train up to 2016, test on 2017-2019, and make the main message 'what still matters once modern commercial climbing is accounted for'.

6. Three course models

Set-up of the first pass (file `03_models.R`): 80/20 train/test split by expedition, tuning by 5-fold cross-validation grouped by expedition, test set used once at the end. Predictors: year, season, sex, age (+ missing flag), citizenship (top 15), role (climber / leader / hired staff), solo, peak (top 10), peak height, 8,000 m peak, peak already climbed, team size, hired staff, staff per member, expedition oxygen, commercial. Metrics: log loss and Brier score (quality of the probabilities), AUC (ranking), accuracy at 0.5.

Model 1: Logistic regression with lasso penalty (units 2 and 3)

What it is	A logistic regression (binomial GLM) with natural splines for year, age and peak height, and an oxygen x role interaction. The lasso penalty shrinks unimportant coefficients to zero.
Why it fits	It is the interpretable benchmark: each coefficient is an odds ratio. Shows variable selection (lasso path) and the bias-variance trade-off through lambda.
Tuning	lambda chosen by grouped 5-fold CV with the 1-SE rule: 40 of 54 terms kept. Next: compare alpha (ridge / elastic net / lasso).

```
glmnet::cv.glmnet(X, y, family = "binomial", alpha = 1, foldid = foldid)
```

Model 2: Classification tree with cost-complexity pruning (unit 5)

What it is	A single CART tree: it splits the climbers into groups with similar success rates.
Why it fits	Easy to show on one slide ('oxygen? -> peak height? -> year?'). Finds interactions automatically. Can use a loss matrix if a missed success and a false alarm should cost differently.
Tuning	Grown deep, then pruned with the 1-SE rule using the grouped CV folds: 63 leaves. For the slide, a smaller tree (about 10 leaves) is easier to read.

```
rpart(success ~ ., data = train, method = "class", control = rpart.control(cp = 0.0005, xval = foldid))
```

Model 3: Random forest (unit 6)

What it is	500 trees, each grown on a bootstrap sample with a random subset of predictors (mtry) at each split; probabilities are averaged.
Why it fits	Usually the most accurate of the three; permutation importance shows which variables matter; OOB error vs CV error is a unit 6 topic.
Tuning	mtry in {2, 4, 7} x min node size in {20, 100, 400} by grouped CV: best mtry = 7, node size 100. The best mtry is at the edge of the grid, so a second grid is needed.

```
ranger(success ~ ., data = train, num.trees = 500, mtry = 7, min.node.size = 100, probability = TRUE)
```

First results on the test set (14,094 climbers, 1,979 expeditions)

Model	Log loss	Brier	AUC	Accuracy
Baseline (always predict 40.7%)	0.674	0.240	0.50	59.9%
Lasso logistic regression	0.570	0.193	0.76	70.0%
Pruned classification tree	0.556	0.186	0.77	73.4%
Random forest	**0.519**	**0.173**	**0.81**	**74.9%**
Random forest + member oxygen (leaky, do not use)	0.468	0.152	0.85	78.0%

Lower log loss and Brier are better; higher AUC is better.

What the first pass shows

- All three models beat the baseline clearly. The random forest is best on every metric.
- The tree beats the lasso, so there are strong interactions (for example oxygen x peak height x year) that a main-effects model misses. This is a good discussion point for the slides.
- Random forest permutation importance: expedition oxygen > peak height > year > peak > staff per member > role > team size > hired staff > citizenship > season. Sex and solo add almost nothing once the rest is included.
- Largest lasso effects: peak height (non-linear), age (declining), expedition oxygen (+), year (+), hired staff with oxygen (+), Cho Oyu (+), Makalu (-).
- The leakage check shows how much a single wrong variable inflates the results.

7. Files and next steps

File (in himalaya/)	What it does
01_describe.R	Loads the three tables and describes every variable -> results_describe.txt
02_prepare.R	Fixes, filters and features -> data/model_data.rds, figures/eda_overview.png
03_models.R	Lasso, tree and random forest with grouped CV -> results_models.txt, results_test_metrics.csv
04_report_figures.R	Extra figures used in this report

- Decide the headline question (recommended: A + B) and switch to the time-based test set.
- Second tuning stage for the random forest (mtry above 7), alpha grid for the elastic net.
- Optional extra models from the course: boosting (unit 7) or a small neural network (unit 8).
- Interpretation: partial dependence plots for year, age, peak height and oxygen; calibration plot; a small readable tree.